

Distributional Semantic Models

Part 4: Elements of matrix algebra

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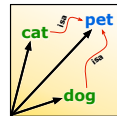
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Outline

Matrix algebra

Roll your own DSM

Matrix multiplication

Association scores & normalization

Geometry

Metrics and norms

Angles and orthogonality

Dimensionality reduction

Orthogonal projection

PCA & SVD

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Matrices and vectors

- ▶ $k \times n$ matrix $\mathbf{M} \in \mathbb{R}^{k \times n}$ is a rectangular array of real numbers

$$\mathbf{M} = \begin{bmatrix} m_{11} & \cdots & m_{1n} \\ \vdots & & \vdots \\ m_{k1} & \cdots & m_{kn} \end{bmatrix}$$

- ▶ Each row $\mathbf{m}_i \in \mathbb{R}^n$ is an n -dimensional vector

$$\mathbf{m}_i = (m_{i1}, m_{i2}, \dots, m_{in})$$

- ▶ Similarly, each column is a k -dimensional vector $\in \mathbb{R}^k$

```
> options(digits=3)
> M <- DSM_TermTerm$M
> M[2, ] # row vector  $\mathbf{m}_2$  for "dog"
> M[, 5] # column vector for "important"
```

Matrices and vectors

- ▶ Vector $\mathbf{x} \in \mathbb{R}^n$ as single-row or single-column matrix
 - ▶ $\mathbf{x} = \mathbf{x}^{TT} = n \times 1$ matrix ("vertical")
 - ▶ $\mathbf{x}^T = 1 \times n$ matrix ("horizontal")
 - ▶ **transposition** operator $\cdot^T$ swaps rows & columns of matrix
- ▶ We need vectors $\mathbf{r} \in \mathbb{R}^k$ and $\mathbf{c} \in \mathbb{R}^n$ of marginal frequencies
- ▶ Notation for cell ij of co-occurrence matrix:
 - ▶ $m_{ij} = O \dots$ observed co-occurrence frequency
 - ▶ $r_i = R \dots$ row marginal (target)
 - ▶ $c_j = C \dots$ column marginal (feature)
 - ▶ $N \dots$ sample size

```
> r <- DSM_TermTerm$rows$f
> c <- DSM_TermTerm$cols$f
> N <- DSM_TermTerm$globals$N
> t(r)      # "horizontal" vector
> t(t(r))   # "vertical" vector
```

Scalar operations

- ▶ **Scalar** operations perform the same transformation on each element of a vector or matrix, e.g.
 - ▶ add / subtract fixed shift $\mu \in \mathbb{R}$
 - ▶ multiply / divide by fixed factor $\sigma \in \mathbb{R}$
 - ▶ apply function ($\log, \sqrt{\cdot}, \dots$) to each element
- ▶ Allows efficient processing of large sets of values
- ▶ Element-wise binary operators on matching vectors / matrices
 - ▶ $\mathbf{x} + \mathbf{y} =$ **vector addition**
 - ▶ $\mathbf{x} \odot \mathbf{y} =$ element-wise multiplication (**Hadamard product**)

```
> log(M + 1) # discounted log frequency weighting
> (M["cause", ] + M["effect", ]) / 2 # centroid vector
```

The outer product

- ▶ Compute matrix $\mathbf{E} \in \mathbb{R}^{k \times n}$ of expected frequencies

$$e_{ij} = \frac{r_i c_j}{N}$$

i.e. $\mathbf{r}[i] * \mathbf{c}[j]$ for each cell ij

- ▶ This is the **outer product** of $\mathbf{r}$ and $\mathbf{c}$

$$\begin{bmatrix} r_1 \\ \vdots \\ r_k \end{bmatrix} \cdot \begin{bmatrix} c_1 & c_2 & \dots & c_n \end{bmatrix} = \begin{bmatrix} r_1 c_1 & r_1 c_2 & \dots & r_1 c_n \\ \vdots & \vdots & & \vdots \\ r_k c_1 & r_k c_2 & \dots & r_k c_n \end{bmatrix}$$

- ▶ The **inner product** of $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is the sum $x_1 y_1 + \dots + x_n y_n$

```
> outer(r, c) / N
```

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Matrix multiplication

$$\begin{bmatrix} a_{ij} \end{bmatrix} = \begin{bmatrix} b_{i1} & \dots & b_{in} \end{bmatrix} \cdot \begin{bmatrix} c_{1j} \\ \vdots \\ c_{nj} \end{bmatrix}$$

$$\mathbf{A} = \mathbf{B} \cdot \mathbf{C}$$

$$(k \times m) = (k \times n) \cdot (n \times m)$$

- ▶ **B** and **C** must be **conformable** (in dimension n)
- ▶ Element a_{ij} is the inner product of the i -th row of **B** and the j -th column of **C**

$$a_{ij} = b_{i1}c_{1j} + \dots + b_{in}c_{nj} = \sum_{t=1}^n b_{it}c_{tj}$$

Some properties of matrix multiplication

Associativity: $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C} =: \mathbf{ABC}$

Distributivity: $\mathbf{A}(\mathbf{B} + \mathbf{B}') = \mathbf{AB} + \mathbf{AB}'$
 $(\mathbf{A} + \mathbf{A}')\mathbf{B} = \mathbf{AB} + \mathbf{A'B}$

Scalar multiplication: $(\lambda\mathbf{A})\mathbf{B} = \mathbf{A}(\lambda\mathbf{B}) = \lambda(\mathbf{AB}) =: \lambda\mathbf{AB}$

- ▶ Not commutative in general: $\mathbf{AB} \neq \mathbf{BA}$
- ▶ The k -dimensional square-diagonal **identity matrix**

$$\mathbf{I}_k := \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \quad \text{with} \quad \mathbf{I}_k \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{I}_n = \mathbf{A}$$

is the **neutral element** of matrix multiplication

Transposition and multiplication

- ▶ The **transpose** $\mathbf{A}^T$ of a matrix **A** swaps rows and columns:

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}^T = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$$

- ▶ Properties of the transpose:
 - ▶ $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$
 - ▶ $(\lambda\mathbf{A})^T = \lambda(\mathbf{A}^T) =: \lambda\mathbf{A}^T$
 - ▶ $(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$ [note the different order of **A** and **B**!]
 - ▶ $\mathbf{I}^T = \mathbf{I}$
- ▶ **A** is called **symmetric** iff $\mathbf{A}^T = \mathbf{A}$
 - ▶ symmetric matrices have many special properties that will become important later (e.g. eigenvalues)

The outer product as matrix multiplication

- ▶ The outer product is a special case of matrix multiplication

$$\mathbf{E} = \frac{1}{N}(\mathbf{r} \cdot \mathbf{c}^T)$$

- ▶ The other special case is the **inner product**

$$\mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

- ▶ NB: $\mathbf{x} \cdot \mathbf{x}$ and $\mathbf{x}^T \cdot \mathbf{x}^T$ are not conformable

three ways to compute the matrix of expected frequencies

```
> E <- outer(r, c) / N
> E <- (r %*% t(c)) / N
> E <- tcrossprod(r, c) / N
> E
```

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Normalizing vectors

- Compute Euclidean norm of vector $\mathbf{x} \in \mathbb{R}^n$:

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$$

- Normalized vector $\|\mathbf{x}_0\|_2 = 1$ by scalar multiplication:

$$\mathbf{x}_0 = \frac{1}{\|\mathbf{x}\|_2} \mathbf{x}$$

```
> x <- S[2, ]
> b <- sqrt(sum(x ^ 2)) # Euclidean norm of x
> x0 <- x / b          # normalized vector
> sqrt(sum(x0 ^ 2))
```

Computing association scores

- Association scores = element-wise combination of $\mathbf{M}$ and $\mathbf{E}$, e.g. for (pointwise) Mutual Information

$$\mathbf{S} = \log_2(\mathbf{M} \oslash \mathbf{E})$$

- $\oslash$ = element-wise division similar to Hadamard product $\odot$
- For sparse AMs such as PPMI, we need to compute $\max\{s_{ij}, 0\}$ for each element of the scored matrix $\mathbf{S}$

```
> log2(M / E)
> S <- pmax(log2(M / E), 0) # not max() !
> S
```

Normalizing matrix rows

- Compute vector $\mathbf{b} \in \mathbb{R}^k$ of norms of row vectors of $\mathbf{S}$
- Can you find an elegant way to multiply each row of $\mathbf{S}$ with the corresponding normalization factor b_i^{-1} ?
- Multiplication with **diagonal matrix** $\mathbf{D}_\mathbf{b}^{-1}$

$$\mathbf{S}_0 = \mathbf{D}_\mathbf{b}^{-1} \cdot \mathbf{S}$$

$$\mathbf{S}_0 = \begin{bmatrix} b_1^{-1} & & \\ & \ddots & \\ & & b_k^{-1} \end{bmatrix} \cdot \begin{bmatrix} s_{11} & \cdots & s_{1n} \\ \vdots & & \vdots \\ s_{k1} & \cdots & s_{kn} \end{bmatrix}$$

- What about multiplication with diagonal matrix on the right?

Normalizing matrix rows

- ▶ Compute vector $\mathbf{b} \in \mathbb{R}^k$ of norms of row vectors of $\mathbf{S}$
- ▶ Can you find an elegant way to multiply each row of $\mathbf{S}$ with the corresponding normalization factor b_i^{-1} ?
- ▶ Multiplication with **diagonal matrix** $\mathbf{D}_b^{-1}$

$$\mathbf{S}_0 = \mathbf{D}_b^{-1} \cdot \mathbf{S}$$

```
> b <- sqrt(rowSums(S^2))
> b <- rowNorms(S, method="euclidean") # more efficient

> S0 <- diag(1 / b) %*% S
> S0 <- scaleMargins(S, rows=(1 / b)) # much more efficient

> S0 <- normalize.rows(S, method="euclidean") # the easy way
```

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Metric: a measure of distance

- ▶ A **metric** is a general measure of the distance $d(\mathbf{u}, \mathbf{v})$ between points $\mathbf{u}$ and $\mathbf{v}$, which satisfies the following axioms:
 - ▶ $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$
 - ▶ $d(\mathbf{u}, \mathbf{v}) > 0$ for $\mathbf{u} \neq \mathbf{v}$
 - ▶ $d(\mathbf{u}, \mathbf{u}) = 0$
 - ▶ $d(\mathbf{u}, \mathbf{w}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{w})$ (**triangle inequality**)
- ▶ Metrics form a very broad class of distance measures, some of which do not fit in well with our geometric intuitions
- ▶ Useful: family of **Minkowski** p -metrics

$$d_p(\mathbf{u}, \mathbf{v}) := (|u_1 - v_1|^p + \dots + |u_n - v_n|^p)^{1/p} \quad p \geq 1$$

$$d_p(\mathbf{u}, \mathbf{v}) := |u_1 - v_1|^p + \dots + |u_n - v_n|^p \quad 0 \leq p < 1$$

with limiting cases

$$d_0(\mathbf{u}, \mathbf{v}) = \#\{i \mid u_i \neq v_i\}$$

$$d_\infty(\mathbf{u}, \mathbf{v}) = \max\{|u_1 - v_1|, \dots, |u_n - v_n|\}$$

Norm: a measure of length

- ▶ A general **norm** $\|\mathbf{u}\|$ for the length of a vector $\mathbf{u}$ must satisfy the following axioms:
 - ▶ $\|\mathbf{u}\| > 0$ for $\mathbf{u} \neq \mathbf{0}$
 - ▶ $\|\lambda \mathbf{u}\| = |\lambda| \cdot \|\mathbf{u}\|$ (**homogeneity**)
 - ▶ $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (**triangle inequality**)
- ▶ Every norm **induces** a metric

$$d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$$

with two desirable properties

- ▶ **translation-invariant**: $d(\mathbf{u} + \mathbf{x}, \mathbf{v} + \mathbf{x}) = d(\mathbf{u}, \mathbf{v})$
- ▶ **scale-invariant**: $d(\lambda \mathbf{u}, \lambda \mathbf{v}) = |\lambda| \cdot d(\mathbf{u}, \mathbf{v})$
- ▶ $d_p(\mathbf{u}, \mathbf{v})$ is induced by the **Minkowski norm** for $p \geq 1$:

$$\|\mathbf{u}\|_p := (|u_1|^p + \dots + |u_n|^p)^{1/p}$$

Norm: a measure of length

- ▶ A generalized **p -norm** $\|\mathbf{u}\|$ for the length of a vector $\mathbf{u}$ satisfies the following relaxed axioms:
 - ▶ $\|\mathbf{u}\| > 0$ for $\mathbf{u} \neq \mathbf{0}$
 - ▶ $\|\lambda \mathbf{u}\| = |\lambda|^p \cdot \|\mathbf{u}\|$ (for some $p > 0$)
 - ▶ $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (**triangle inequality**)
- ▶ A p -norm **induces** a metric $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$ that is not scale-invariant (unless $p = 1$), but has properties
 - ▶ **translation-invariant**: $d(\mathbf{u} + \mathbf{x}, \mathbf{v} + \mathbf{x}) = d(\mathbf{u}, \mathbf{v})$
 - ▶ **scaling behaviour**: $d(\lambda \mathbf{u}, \lambda \mathbf{v}) = |\lambda|^p \cdot d(\mathbf{u}, \mathbf{v})$
- ▶ $d_p(\mathbf{u}, \mathbf{v})$ is induced by the **Minkowski p -norm** for $0 < p < 1$:

$$\|\mathbf{u}\|_p := |u_1|^p + \dots + |u_n|^p$$

☞ the Hamming metric d_0 is not norm-induced because $\|\cdot\|_0$ is not a p -norm (note that $\|\lambda \mathbf{u}\|_0 = \|\mathbf{u}\|_0$ for $\lambda \neq 0$)

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Euclidean norm & inner product

- ▶ The Euclidean norm $\|\mathbf{u}\|_2 = \sqrt{\mathbf{u}^T \mathbf{u}}$ is special because it can be derived from the **inner product**:

$$\mathbf{x}^T \mathbf{y} = x_1 y_1 + \dots + x_n y_n$$

- ▶ The inner product is a **positive definite** and **symmetric bilinear form** with an important geometric interpretation:

$$\cos \phi = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\|_2 \cdot \|\mathbf{v}\|_2}$$

for the **angle** ϕ between vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$

- ▶ the value $\cos \phi$ is known as the **cosine similarity** measure
- ▶ In particular, $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** iff $\mathbf{u}^T \mathbf{v} = 0$

Cosine similarity in R

- ▶ Cosine similarities can be computed very efficiently if vectors are pre-normalized, so that $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$

☞ just need all inner products $\mathbf{m}_i^T \mathbf{m}_j$ between row vectors of $\mathbf{M}$

$$\mathbf{M} \cdot \mathbf{M}^T = \begin{bmatrix} \dots & \mathbf{m}_1 & \dots \\ \dots & \mathbf{m}_2 & \dots \\ \dots & \mathbf{m}_k & \dots \end{bmatrix} \cdot \begin{bmatrix} \vdots & \vdots & \vdots \\ \mathbf{m}_1 & \mathbf{m}_2 & \mathbf{m}_k \\ \vdots & \vdots & \vdots \end{bmatrix}$$

$$\Rightarrow (\mathbf{M} \cdot \mathbf{M}^T)_{ij} = \mathbf{m}_i^T \mathbf{m}_j$$

```
# cosine similarities for row-normalized matrix:
> sim <- tcrossprod(S0)
> angles <- acos(pmin(sim, 1)) * (180 / pi)
```

Euclidean distance or cosine similarity?

- Proof that Euclidean distance and cosine similarity are equivalent looks much simpler in matrix algebra
- Assuming that $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$, we have:

$$\begin{aligned} d_2(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{(\mathbf{u} - \mathbf{v})^T (\mathbf{u} - \mathbf{v})} \\ &= \sqrt{\mathbf{u}^T \mathbf{u} + \mathbf{v}^T \mathbf{v} - 2 \mathbf{u}^T \mathbf{v}} \\ &= \sqrt{\|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2 \mathbf{u}^T \mathbf{v}} \\ &= \sqrt{2 - 2 \cos \phi} \end{aligned}$$

☞ $d_2(\mathbf{u}, \mathbf{v})$ is a monotonically increasing function of ϕ

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Linear subspace & basis

- A linear **subspace** $B \subseteq \mathbb{R}^n$ of rank $r \leq n$ is spanned by a set of r linearly independent basis vectors

$$B = \{\mathbf{b}_1, \dots, \mathbf{b}_r\}$$

- Every point $\mathbf{u}$ in the subspace is a unique linear combination of the basis vectors

$$\mathbf{u} = x_1 \mathbf{b}_1 + \dots + x_r \mathbf{b}_r$$

with coordinate vector $\mathbf{x} \in \mathbb{R}^r$

- Basis matrix $\mathbf{V} \in \mathbb{R}^{n \times r}$ with column vectors $\mathbf{b}_i$:

$$\mathbf{u} = \mathbf{V}\mathbf{x}$$

Linear subspace & basis

- Basis matrix $\mathbf{V} \in \mathbb{R}^{n \times r}$ with column vectors $\mathbf{b}_i$:

$$\mathbf{u} = x_1 \mathbf{b}_1 + \dots + x_r \mathbf{b}_r = \mathbf{V}\mathbf{x}$$

$$\begin{bmatrix} x_1 b_{11} + \dots + x_r b_{r1} \\ x_1 b_{12} + \dots + x_r b_{r2} \\ \vdots \\ x_1 b_{1n} + \dots + x_r b_{rn} \end{bmatrix} = \begin{bmatrix} b_{11} & \dots & b_{r1} \\ b_{12} & \dots & b_{r2} \\ \vdots & & \vdots \\ b_{1n} & \dots & b_{rn} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_r \end{bmatrix}$$

$$\begin{matrix} \mathbf{u} \\ (n \times 1) \end{matrix} = \begin{matrix} \mathbf{V} \\ (n \times r) \end{matrix} \cdot \begin{matrix} \mathbf{x} \\ (r \times 1) \end{matrix}$$

Orthonormal basis

- Particularly convenient with orthonormal basis:

$$\begin{aligned}\|\mathbf{b}_i\|_2 &= 1 \\ \mathbf{b}_i^T \mathbf{b}_j &= 0 \quad \text{for } i \neq j\end{aligned}$$

- Corresponding basis matrix $\mathbf{V}$ is (column)-**orthogonal**

$$\mathbf{V}^T \mathbf{V} = \mathbf{I}_r$$

and defines a **Cartesian coordinate system** in the subspace

The mathematics of projections

- 1-d subspace spanned by basis vector $\|\mathbf{b}\|_2 = 1$
- For any point $\mathbf{u}$, we have

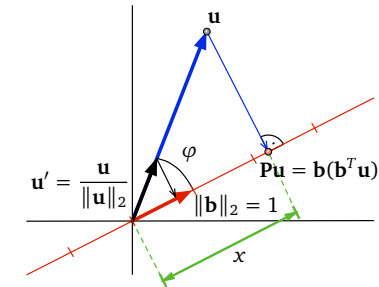
$$\cos \varphi = \frac{\mathbf{b}^T \mathbf{u}}{\|\mathbf{b}\|_2 \cdot \|\mathbf{u}\|_2} = \frac{\mathbf{b}^T \mathbf{u}}{\|\mathbf{u}\|_2}$$

- Trigonometry: coordinate of point on the line is $x = \|\mathbf{u}\|_2 \cdot \cos \varphi = \mathbf{b}^T \mathbf{u}$

- The projected point in original space is then given by

$$\mathbf{b} \cdot x = \mathbf{b}(\mathbf{b}^T \mathbf{u}) = (\mathbf{b}\mathbf{b}^T) \mathbf{u} = \mathbf{P}\mathbf{u}$$

where $\mathbf{P}$ is a **projection matrix** of rank 1



The mathematics of projections

- For an orthogonal basis matrix $\mathbf{V}$ with columns $\mathbf{b}_1, \dots, \mathbf{b}_r$, the projection into the rank- r subspace B is given by

$$\mathbf{P}\mathbf{u} = \left(\sum_{i=1}^r \mathbf{b}_i \mathbf{b}_i^T \right) \mathbf{u} = \mathbf{V}\mathbf{V}^T \mathbf{u}$$

and its subspace coordinates are $\mathbf{x} = \mathbf{V}^T \mathbf{u}$

- Projection can be seen as decomposition into the projected vector and its orthogonal complement

$$\mathbf{u} = \mathbf{P}\mathbf{u} + (\mathbf{u} - \mathbf{P}\mathbf{u}) = \mathbf{P}\mathbf{u} + (\mathbf{I} - \mathbf{P})\mathbf{u} = \mathbf{P}\mathbf{u} + \mathbf{Q}\mathbf{u}$$

- Because of orthogonality, this also applies to the squared Euclidean norm (according to the Pythagorean theorem)

$$\|\mathbf{u}\|^2 = \|\mathbf{P}\mathbf{u}\|^2 + \|\mathbf{Q}\mathbf{u}\|^2$$

Aside: the matrix cross-product

- We already know that the (transpose) cross-product $\mathbf{M}\mathbf{M}^T$ computes all inner products between the row vectors of $\mathbf{M}$
- But $\mathbf{V}\mathbf{V}^T$ it can also be understood as a **superposition** of the **outer products** of the columns of $\mathbf{V}$ with themselves

$$\mathbf{V}\mathbf{V}^T = \begin{bmatrix} b_{11} \\ \vdots \\ b_{1n} \end{bmatrix} \cdot [b_{11} \cdots b_{1n}] + \dots + \begin{bmatrix} b_{r1} \\ \vdots \\ b_{rn} \end{bmatrix} \cdot [b_{r1} \cdots b_{rn}]$$

Projections in R

```
# column basis vector for "animal" subspace
> b <- t(t(c(1, 1, 1, 1, .5, 0, 0)))
> b <- normalize.cols(b) # basis vectors must be normalized

> (x <- M %%% b) # projection of data points into subspace coordinates
> x %%% t(b)      # projected points in original space
> tcrossprod(x, b) # outer() only works for plain vectors

> P <- b %%% t(b) # projection operator
> P - t(P)        # note that P is symmetric
> M %%% P         # projected points in original space
```

Optimal projections and subspaces

- Orthogonal decomposition of squared distances btw vectors

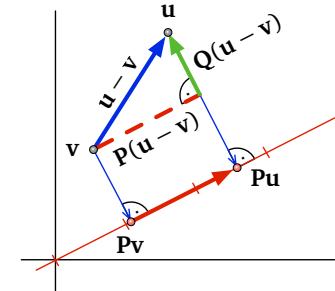
$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{P}\mathbf{u} - \mathbf{P}\mathbf{v}\|^2 + \|\mathbf{Q}\mathbf{u} - \mathbf{Q}\mathbf{v}\|^2$$

- Define projection **loss** as difference btw squared distances

$$\begin{aligned} & \left| \|\mathbf{P}(\mathbf{u} - \mathbf{v})\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 \right| \\ &= \|\mathbf{u} - \mathbf{v}\|^2 - \|\mathbf{P}(\mathbf{u} - \mathbf{v})\|^2 \\ &= \|\mathbf{Q}(\mathbf{u} - \mathbf{v})\|^2 \end{aligned}$$

- Projection quality measure:

$$R^2 = \frac{\|\mathbf{P}(\mathbf{u} - \mathbf{v})\|^2}{\|\mathbf{u} - \mathbf{v}\|^2}$$



Optimal projections and subspaces

- Optimal subspace maximises R^2 across a data set $\mathbf{M}$, which is now specified in terms of row vectors $\mathbf{m}_i^T$:

$$\begin{aligned} \mathbf{x}_i^T &= \mathbf{m}_i^T \mathbf{V} & \mathbf{m}_i^T \mathbf{P} &= \mathbf{m}_i^T \mathbf{V} \mathbf{V}^T \\ \mathbf{X} &= \mathbf{M} \mathbf{V} & \mathbf{M} \mathbf{P} &= \mathbf{M} \mathbf{V} \mathbf{V}^T \end{aligned}$$

- We will now show that the overall projection quality is

$$R^2 = \frac{\sum_{i=1}^k \|\mathbf{m}_i^T \mathbf{P}\|^2}{\sum_{i=1}^k \|\mathbf{m}_i^T\|^2} = \frac{\|\mathbf{M} \mathbf{P}\|_F^2}{\|\mathbf{M}\|_F^2}$$

with the (squared) **Frobenius norm**

$$\|\mathbf{M}\|_F^2 = \sum_{ij} (m_{ij})^2 = \sum_{i=1}^k \|\mathbf{m}_i\|^2$$

Optimal projections and subspaces

- For a **centered** data set with $\sum_i \mathbf{m}_i = \mathbf{0}$, the Frobenius norm corresponds to the average (squared) distance between points

$$\begin{aligned} \sum_{i,j=1}^k \|\mathbf{m}_i - \mathbf{m}_j\|^2 &= \sum_{i,j=1}^k (\mathbf{m}_i - \mathbf{m}_j)^T (\mathbf{m}_i - \mathbf{m}_j) \\ &= \sum_{i,j=1}^k (\|\mathbf{m}_i\|^2 + \|\mathbf{m}_j\|^2 - 2\mathbf{m}_i^T \mathbf{m}_j) \\ &= \sum_{j=1}^k \|\mathbf{M}\|_F^2 + \sum_{i=1}^k \|\mathbf{M}\|_F^2 - 2 \sum_{i=1}^k \mathbf{m}_i^T \underbrace{\left(\sum_{j=1}^k \mathbf{m}_j \right)}_0 \\ &= 2k \cdot \|\mathbf{M}\|_F^2 \end{aligned}$$

- Similarly for the overall projection loss and quality R^2 :

$$R^2 = \frac{\sum_{i,j=1}^k \|\mathbf{P}(\mathbf{m}_i - \mathbf{m}_j)\|^2}{\sum_{i,j=1}^k \|\mathbf{m}_i - \mathbf{m}_j\|^2} = \frac{2k \cdot \|\mathbf{M} \mathbf{P}\|_F^2}{2k \cdot \|\mathbf{M}\|_F^2}$$

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Singular value decomposition

- Fundamental result of matrix algebra: **singular value decomposition (SVD)** factorises any matrix $\mathbf{M}$ into

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal and $\mathbf{\Sigma}$ is a diagonal matrix of **singular values** $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m > 0$

$$\begin{bmatrix} n \\ k \end{bmatrix} \mathbf{M} = \begin{bmatrix} m \\ k \end{bmatrix} \mathbf{U} \cdot \begin{bmatrix} \sigma_1 & m \\ m & \ddots \\ & \mathbf{\Sigma} & \sigma_m \end{bmatrix} \cdot \begin{bmatrix} n \\ m \end{bmatrix} \mathbf{V}^T$$

Singular value decomposition

- $m \leq \min\{k, n\}$ is the inherent dimensionality (**rank**) of $\mathbf{M}$
- Columns $\mathbf{a}_i$ of $\mathbf{U}$ are called left singular vectors, columns $\mathbf{b}_i$ of $\mathbf{V}$ (= rows of $\mathbf{V}^T$) are right singular vectors
- Recall the “outer product” view of matrix multiplication:

$$\mathbf{U}\mathbf{V}^T = \sum_{i=1}^m \mathbf{a}_i \mathbf{b}_i^T$$

- Hence the SVD corresponds to a sum of rank-1 components

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_{i=1}^m \sigma_i \mathbf{a}_i \mathbf{b}_i^T$$

Singular value decomposition

- Key property of SVD: the first r components give the best rank- r approximation to $\mathbf{M}$ with respect to the Frobenius norm, i.e. they minimize the loss

$$\|\mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T - \mathbf{M}\|_F^2 = \|\mathbf{M}_r - \mathbf{M}\|_F^2$$

Truncated SVD

- $\mathbf{U}_r, \mathbf{V}_r$ = first r columns of $\mathbf{U}, \mathbf{V}$
- $\mathbf{\Sigma}_r$ = diagonal matrix of first r singular values
- It can be shown that

$$\|\mathbf{M}\|_F^2 = \sum_{i=1}^m \sigma_i^2 \quad \text{and} \quad \|\mathbf{M}_r\|_F^2 = \sum_{i=1}^r \sigma_i^2$$

SVD dimensionality reduction

- Columns of $\mathbf{V}_r$ form an orthogonal basis of the optimal rank- r subspace because

$$\mathbf{M}\mathbf{P} = \mathbf{M}\mathbf{V}_r\mathbf{V}_r^T = \mathbf{U}\underbrace{\boldsymbol{\Sigma}\mathbf{V}_r^T\mathbf{V}_r}_{=\mathbf{I}_r}\mathbf{V}_r^T = \mathbf{U}_r\boldsymbol{\Sigma}_r\mathbf{V}_r^T = \mathbf{M}_r$$

- Dimensionality reduction** uses the subspace coordinates

$$\mathbf{M}\mathbf{V}_r = \mathbf{U}_r\boldsymbol{\Sigma}_r$$

- If $\mathbf{M}$ is centered, this also gives the best possible preservation of pairwise distances → **principal component analysis (PCA)**
 - but centering is usually omitted in order to maintain sparseness, so SVD preserves vector lengths rather than distances

Scaling SVD dimensions

- Singular values σ_i can be seen as weighting of the latent dimensions, which determines their contribution to

$$\|\mathbf{M}\mathbf{V}_r\|_F = \sigma_1^2 + \dots + \sigma_r^2$$

- Weighting adjusted by **power scaling** of the singular values:

$$\mathbf{U}_r\boldsymbol{\Sigma}_r^p = \begin{bmatrix} \vdots & & \vdots \\ \sigma_1^p \mathbf{a}_1 & \cdots & \sigma_r^p \mathbf{a}_r \\ \vdots & & \vdots \end{bmatrix}$$

- $p = 1$: normal SVD projection
- $p = 0$: dimension weights equalized
- $p = 2$: more weight given to first latent dimensions
- Other weighting schemes possible (e.g. skip first dimensions)

SVD projection in R

```
> fact <- svd(S0)      # SVD decomposition of S0
> round(fact$u, 3)      # left singular vectors (columns) = U
> round(fact$v, 3)      # right singular vectors (columns) = V
> round(fact$d, 3)      # singular values = diagonal of Σ
# note that S0 has effective rank 6 because σ7 ≈ 0
> barplot(fact$d ~ 2)   # R² contributions

> r <- 2                # truncated rank-2 SVD
> (U.r <- fact$u[, 1:r])
> (Sigma.r <- diag(fact$d[1:r], nrow=r))
> (V.r <- fact$v[, 1:r])
```

SVD projection in R

```
> (X.r <- S0 %*% V.r) # project into latent coordinates
> U.r %*% Sigma.r      # same result
> scaleMargins(U.r, cols=fact$d[1:r]) # the wordspace way

> rownames(X.r) <- rownames(S0)      # NB: keep row labels

> S0r <- U.r %*% Sigma.r %*% t(V.r)  # rank-2 matrix approx.
> round(S0r, 3)
# compare with S0: where are the differences?

> round(X.r %*% t(V.r), 3)           # same result
```

- see example code for comparison against PCA with centering